

ICT Health and Safety Program

Part II. Lab Users

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Chemical Safety

- **Part I: Introduction to Hazards**
 - Recognize hazards
 - Reading SDS and interpreting hazard symbols
- **Part II: Assess and Minimize the risks**
 - Safe chemical handling
 - Bonding and grounding; Transferring to smaller containers
 - Storage and waste disposal
- **Part III: Prepare for Emergencies**

Chemical Safety – Recognize Hazard

- **Read the label and Safety Data Sheet**
 - Uses global harmonized system (GHS) hazard classification
 - Category numbers based on severity
 - “4” least hazardous; “1” most hazardous
 - Signal words
 - “Warning” is for less severe; “Danger” is for more severe
 - Includes pictograms (requirement for flammables categories 1 – 3)
 - absence doesn’t indicate no hazard.

Chemical Safety – GHS Hazard Classification

Common chemicals at ICT



Acetone



Trichloroethylene



Mineral spirits

Chemical Safety – GHS Hazard Classification

■ GHS Pictograms

Health Hazard



- Carcinogen
- Mutagenicity
- Reproductive Toxicity
- Respiratory Sensitizer
- Target Organ Toxicity
- Aspiration Toxicity

Skull and Crossbones



- Acute Toxicity (fatal or toxic)

Exclamation Mark



- Irritant (skin and eye)
- Skin Sensitizer
- Acute Toxicity (harmful)
- Narcotic Effects
- Respiratory Tract Irritant
- Hazardous to Ozone Layer (Non-Mandatory)

Chemical Safety – GHS Hazard Classification

■ GHS Pictograms

Gas Cylinder



- Gases Under Pressure

Flame



- Flammables
- Pyrophorics
- Self-Heating
- Emits Flammable Gas
- Self-Reactives
- Organic Peroxides

Corrosion



- Skin Corrosion/
Burns
- Eye Damage
- Corrosive to Metals

Chemical Safety – Safety Data Sheets

- Familiarize with sections **1, 2, 4, 7, and 8.**
- Section **1**: Manufacturer, date, emergency contact
- Section **2**: GHS Classification

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Skin Corrosion/irritation	Category 2
Serious Eye Damage/Eye Irritation	Category 2
Skin Sensitization	Category 1
Germ Cell Mutagenicity	Category 2
Carcinogenicity	Category 1A
Specific target organ toxicity (single exposure)	Category 3
Target Organs - Central nervous system (CNS).	
Specific target organ toxicity - (repeated exposure)	Category 2
Target Organs - Kidney, Liver, Heart, spleen, Blood.	

Chemical Safety – Safety Data Sheets

■ Section 4: First Aid measures

4. First-aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation

Move to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

Chemical Safety – Safety Data Sheets

■ Section 7: Storage Conditions

7. Handling and storage

Handling

Wear personal protective equipment. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe vapors or spray mist. Do not ingest.

Storage

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from light. Do not store in aluminum containers.

■ Section 8: OSHA exposure limits and PPE

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Trichloroethylene	TWA: 10 ppm STEL: 25 ppm	(Vacated) TWA: 50 ppm (Vacated) TWA: 270 mg/m ³ Ceiling: 200 ppm (Vacated) STEL: 200 ppm (Vacated) STEL: 1080 mg/m ³ TWA: 100 ppm	IDLH: 1000 ppm	TWA: 100 ppm TWA: 535 mg/m ³ STEL: 200 ppm STEL: 1080 mg/m ³

Chemical Safety – II. Assess the Risks

■ Laboratory Risk Assessment Tool (LabRAT)

Category	Score (0 – 5)	Calculated Sum Outcome
Chemical Volume Total		< 15 : Low The procedure can be performed with routine precautions
Hazard Recognition		
Special Hazards		15 – 25 : Moderate The procedure can be performed with attention given to specific hazards. Supervision is recommended
Process Conditions		
Explosive Hazard		26 – 30 : High The procedure should be revised to reduce the risk. High level attention must be given to all hazards. High level, continuous supervision is mandatory
Other Hazards		
Procedure Description		30 : Extreme The procedure MUST be revised to reduce the risk
Personnel Training		
Ventilation		
Equipment Maintenance		
Calculate Sum		

- Category scoring is based on a 0 – 5 scale, with 0 being *NOT APPLICABLE* and 5 being *EXTREME*.

Chemical Safety – III. Minimize the Risks

- **Engineering controls – Fume hoods**
 - Allow adequate air movement
 - Elevate large pieces of equipment
 - Do not place many items in the fume hood
 - Reduce vapor turbulence
 - Work at least 6” into the fume hood
 - Limit walking in front of the fume hood
 - Never put your head inside the fume hood
 - Lower the sash when not actively working



Chemical Safety – III. Minimize the Risks

- **Administrative controls – Chemical handling**
 - Attach dates to each chemical received – RE
 - Before transferring chemicals to smaller containers ensure
 - Have proper PPE – safety glasses, flame resistant lab coat (blue), appropriate gloves
 - Use the right container (correct label or color)
 - Dispensing containers are properly bonded – if not contact RE



Chemical Safety – III. Minimize the Risks

■ Administrative controls – Chemical storage

- Only store in labeled containers in good condition
- Group by compatibility.
- Do not change the storage location of chemical
RE approval
- Store in the Flammable cabinet in solvent
room



Chemical Safety – III. Minimize the Risks

- **Personal Protective Equipment (PPE)**
 - Appropriate PPE is required before working with chemicals
 - Safety glasses – required all the time, must meet ANSI z87+
 - Flame resistant lab coat (blue) when working with flammable liquids
 - Cotton lab coats (white) when working with NON flammable liquids
 - Appropriate gloves
 - Nitrile gloves for mineral spirits and acetone
 - Polyvinyl Alcohol (PVA) gloves for TCE

Chemical Safety – IV. Emergency Preparedness

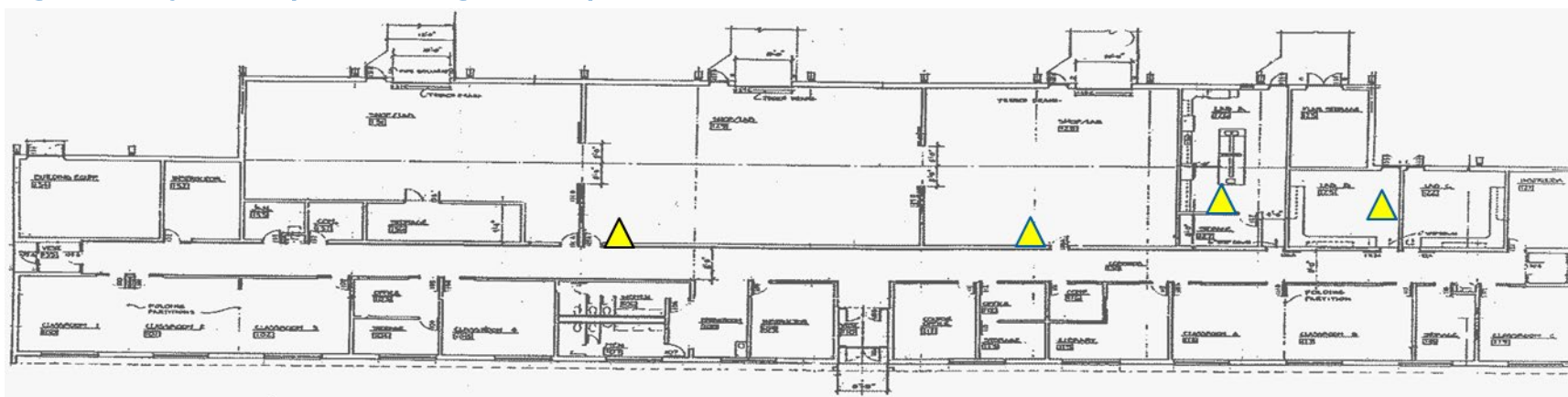
■ Simple chemical spills

- Review SDS section 6: Accidental release measures
- Shut off any ignition sources
- Have appropriate PPE – safety glasses, appropriate gloves, lab coat
- Use absorbent pads or spill kit in solvent room
- Provide adequate ventilation

■ Complicated chemical spills: Confine area if possible and call 911. Provide responders with SDS

Chemical Safety – IV. Emergency Preparedness

- **Locations of eyewash and safety shower stations**
 - Lab A (binder) – South Wall
 - Lab B (Volumetrics) – East Wall
 - Highbay B (near servo-room) – South Wall
 - Highbay C (mixing lab) – South Wall





Other Lab Policies – Equipment Use

- Schedule and obtain training from Research Engineers for every equipment
- Must use clustermarket to reserve equipment

My Lab Activities



✓	7 minutes ago	Equipment	TADSR		28/08/2023 10:00	28/08/2023 10:15	Single	Booked	
✓	7 minutes ago	Equipment	TADSR		28/08/2023 10:00	28/08/2023 18:00	Single	Booked	
✓	in 2 days	Equipment	TADSR		30/08/2023 08:00	30/08/2023 17:00	Single	Booked	

Other Lab Policies – Golf Car Use

- **Complete the Campus Unlicensed Motorized Vehicle Operators Brief**
- **Complete the Departmental Driver Approval form**
- **Use the electric golf car to move items (aggregates, batches, etc.) between MPF and main building or to ATLAS yard**

Other Lab Policies – Use Violations

- **The following violations will lead to suspensions**
 - Violations of PPE requirements – safety toe shoes, safety glasses, sturdy clothes, and long pants.
 - Failure to clean equipment and/or workstations
 - Food and/or drinks in non-approved areas
 - Use of headphones / ear buds
 - Use of equipment without reservations on Clustermarket
 - Use of equipment without proper training

Other Lab Policies – Complacency

■ What's complacency?

- It's a mindset where you become comfortable with an existing situation and stop looking for potential hazards
- An example is when you go on autopilot during your commute
 - If nothing goes wrong, you might not even remember the drive
 - If something goes wrong, you might be so checked out that you don't see the danger in time to avoid it.

Other Lab Policies – Complacency

■ Dangers of complacency

- Complacency is a **silent killer**. The severity of the hazard is invisible before things go wrong
 - First, you gain a **false sense of security** about the **risks** when you perform dangerous tasks repeatedly without incident
 - This false sense of security can also result in you **allowing your mind to wander**. You go through the motions of your task while paying less and less attention
 - Being **less attentive** can **lead to mistakes**, missed steps, or sloppily executed safety protocols

Other Lab Policies – Complacency

- **How to combat complacency**
 - Shake up safety routines and put safe work practices in everyone's mind
 - A popular approach is using toolbox talks. These are short informal sessions (weekly or daily) relevant to specific tasks or existing safety concerns.
 - Frequent refresher safety trainings
 - Frequent safety evaluations

Remember

“Your safety is not a permanent trait, rather it is something you have to earn daily”



THANK YOU

Any Questions?

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